INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Introduction to Cryptography

Lab Session 4: Block ciphers I *March 12, 2026*

1. Programming exercises

Please solve the following exercises in pairs. You can use C/C++, C#, Java or Python to develop your source code, but use only one of them to do all the exercises. **Upload screenshots of your source code and screenshots of each function running before the session ends**

1. Design and implement a function that receives as a parameter a binary string of arbitrary length and encode that string in base 64. Also design a function to reverse the process, i.e. this function must take a string in base 64 and decode it to a binary string. You can use a function of the programming language of your choice to do this.
2. Find a cryptographic library in your favorite programming language that still let you use DES and do the following:
   a) Design and implement a function to generate a random key for DES. Use a secure mechanism for randomness. Encode your key in base 64 and print it.
   b) Design and implement a function that receives as a parameter a key and a filename of a textfile. Encipher this file using DES and the key that you previously generate. The ciphertext must be store in a file in base64. The filename of the plaintext and the filename of the ciphertext must be given by the user. The size of your textfile must be greater than 100kb.
   c) Design and implement a function that receives as a parameter a key and a filename of a ciphertext. Decipher this file using DES and the key.
3. Do a program to test your functions. Please consider that your program, must be able to do only one action: key generation, encipher or decipher.
4. Test your program with textfiles of different sizes, for example 1kb, 500kb, 1Mb.
5. See if your program can encipher your source code.

2. Testing your implementation

You must test your implementation in pairs as follows :

1. One student will play the role of Alice and the other one will be Bob
2. Then Alice must generate at random key for DES K. Encode this key en base64, print it and share it with Bob.
3. Alice will encipher with DES a textfile using the key previously generated. Once she has the ciphertext in base64, she must share the file of the ciphertext with Bob.
4. Bob must decipher the ciphertext, using the key that Alice shared with him.